

TTN hacks

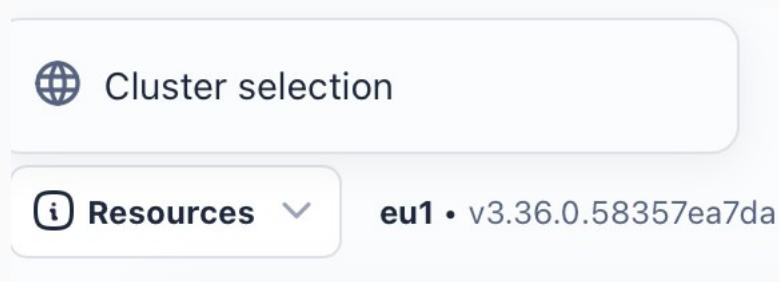
(what went wrong yesterday)

- **TTN Clusters** are regional groups of servers that handle the traffic for The Things Stack. Instead of one giant global server, the network is divided into geographical zones to keep things fast and reliable.
- There are three clusters in TTN:
 - eu1: Europe (Dublin)
 - nam1: North America (California)
 - au1: Australia (Sydney)

- **Always make sure your Gateway and your Application are registered in the same cluster.** If your gateway is in eu1 but your device is registered in nam1, your messages have to travel across the Atlantic Ocean twice, which can cause "downlink" messages to arrive too late for your device to hear them!
- This was the issue we faced! **The gateway was registered in the Australia cluster while the applications were created in the European one.**

■ Solution:

- We have registered the gateway in the European cluster
- Please move to the European cluster yourself (bottom left of the TTN page)



- Create a new application now that you are in the European cluster.

Orphaned devices

- Second issue:

Register end device



Submit failed

An end device with JoinEUI 526973696E674846 and DevEUI 2CF7F1207200054E is already registered as dipdipd1pd1p in application nic1a11112222

[Show details](#)

- If a device has been already registered in a TTN application, it cannot be registered in another application unless it is first deleted from the original application.

Orphaned devices

- Solution:
- If the device is "orphaned," you can bypass the duplicate error by changing the JoinEUI (also known as AppEUI).
- TTN checks for the uniqueness of the pair (DevEUI + JoinEUI). Even if the DevEUI is locked, changing the JoinEUI creates a "new" unique combination.
- For Programmable Devices: **Change the JoinEUI in your device's firmware to a random 64-bit hex string (e.g., 00000000000000001). Register the device in your TTN console using this new JoinEUI.**

Orphaned devices

- Solution:
- For the Seeed Studio LoRa-E5 (now often called the Wio-E5), you can modify the configuration using AT commands through a serial monitor.
- Set the New JoinEUI: Use the following syntax:
 - AT+ID=APPEUI, "000000000000000001"
- In the Day3 directory you will find the file *nicla-ttn-lora-hack.ino* that implements this hack!

- In summary:
- Change your cluster to Europe and create a new application.
- Use the new Arduino code that gives a new JoinEUI address to the device so you can use it again even if it's registered in another application.

Happy LoRaWAN!